Nikhil Bhatt

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Professional Summary

Computer Science Engineering student and Full-Stack Developer experienced in building web applications, real-time distributed systems, and developer tools using **Java, JavaScript, React.js, Node.js, Express.js, and SQL/NoSQL databases**. Built a CRDT-based collaborative code editor, developed secure e-commerce workflows, and published an open-source backend generator on npm. Familiar with **LLM, Retrieval-Augmented Generation (RAG), embeddings, vector databases, and semantic search**, with an interest in applying AI-assisted workflows across software development, testing, and deployment. Strong foundation in REST APIs, automated testing, CI/CD concepts, and Docker-based deployment.

Key Skills & Highlights

- **Languages:** Java, JavaScript (ES6+), SQL
- **Backend & APIs:** Node.js, Express.js, REST API Design, JWT Authentication, Socket.IO, WebSockets
- **AI & GenAI:** LLM Fundamentals, Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases, Semantic Search, AI-Assisted Development
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS, HTML5, CSS3, GSAP, Three.js
- **Databases:** MongoDB, MySQL, Database Schema Design, SQL Queries
- **DevOps, Testing & Tools:** Docker, Git, GitHub, Postman, npm, CI/CD Concepts, E2E Testing, Vercel, Render, Netlify
- **Core Concepts:** Data Structures & Algorithms, OOP, System Design Fundamentals, CRDT (Yjs), Microservices Basics
- **Mobile:** React Native

Projects

Realtime Collaborative Code Editor React.js, Monaco Editor, Node.js, Socket.IO, Yjs, Docker

GitHub | *Live Demo*

- Engineered a real-time collaborative code editor supporting **multiple concurrent users** editing the same document using React, Monaco Editor, Node.js, and Socket.IO.
- Implemented **CRDT-based synchronization** with Yjs to manage concurrent edits and maintain consistent shared document state.
- Built a low-latency **WebSocket communication layer** for real-time document updates and collaboration.
- Implemented live user-presence tracking through Yjs awareness state for multi-user session visibility.
- Load-tested the Socket.IO layer with **1,000 concurrent connections**, achieving **1,000/1,000 successful connections** locally.
- Containerized the application using a **Docker multi-stage build** to support consistent packaging and deployment.

Node.js Backend Generator CLI Node.js, Express.js, JavaScript, npm, CLI, E2E Testing

GitHub | *npm Package*

- Built and **published an open-source CLI tool** (*nikhilbhatt-dev*) on npm that scaffolds backend projects through a single `npx` **command**.
- Designed template-based project generation with recursive file copying, destination-path validation, and controlled environment-file filtering.
- Developed **automated end-to-end tests** covering project generation, error handling, package validation, and dependency installation.
- Validated the published package through a live `npx` workflow, demonstrating reproducible setup and developer-workflow automation.

E-Commerce Full-Stack Website React.js, Node.js, Express.js, MongoDB, Stripe, JWT, Redux, Nodemailer, Cloudinary

GitHub | *Live Demo*

- Developed a full-stack e-commerce platform featuring **JWT authentication**, **Stripe payment integration**, newsletter subscriptions, and an integrated blog module.
- Designed and implemented **RESTful APIs** for authentication, product management, order processing, blog content, and newsletter subscriptions.
- Built an administrative dashboard for managing products, orders, users, blogs, and site content.
- Improved frontend loading behavior using **lazy loading, code splitting, and skeleton loading**.

Achievements & Certifications

- Published **nikhilbhatt-dev**, a reusable Node.js backend generator CLI, on **npm**.
- **Responsive Web Design Developer Certification** — freeCodeCamp
- **AI Agents Course** — Google & Kaggle

Education

Birla Institute of Applied Science, Bhimtal 2023 – Present

Bachelor of Technology in Computer Science Engineering

S.V.M. Inter College, Kashipur 2022 – 2023

Senior Secondary (Class XII)